Lesson 29: Expressing Your Results and Conducting Your Analyses

English Stage 4 (Psychology) · Unit 4: Statistics in Research Reports

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 12.3 Expressing Your Results; 12.4 Conducting Your Analyses; 12.5 Descriptive Statistics (Summary).

In this lesson

You will learn how researchers prepare raw data, handle outliers, separate planned from exploratory analyses, and present descriptive statistics in APA style in sentences, figures and tables. The language focus is the core skill of a results section: reporting statistics in words and numbers.

Learning goals

After this lesson you should be able to:

- Report means, standard deviations and correlations in APA style, in the narrative and in parentheses, with parallel construction.
- Apply the APA guidelines for graphs, captions and tables.
- Explain how raw data are prepared and how outliers and exclusions are handled and reported.
- Distinguish planned from exploratory analyses.

Key language

Term or phrase	Meaning	Example
raw data	data in the form in which they were originally collected	completed paper-and-pencil questionnaires
data file	a computer file that contains data formatted for statistical analysis	one row per participant, one column per variable
error bars / standard error	bars showing variability around a mean, usually one standard error (the *SD* divided by the square root of the sample size)	Error bars represent standard errors.
caption	"*Figure 1.*" (italic) plus a brief description of the figure	*Figure 1.* Reaction times of the control versus experimental group.
correlation matrix	a table that shows the correlations among several variables	working memory and executive function: .96
outlier	an extreme score far removed from the rest	a reaction time of 3 minutes
planned analysis	an analysis that tests a relationship expected in your hypothesis	means, a bar graph and Cohen's *d*

Term or phrase	Meaning	Example
exploratory analysis	an analysis undertaken without an existing hypothesis	Bem's "fishing expedition"
parallel construction	expressing similar results in similar ways	The treatment group had a mean of 23.40 (*SD* = 9.33), while the control group had a mean of 20.87 (*SD* = 8.45).
set aside (data)	keep excluded data separately instead of deleting them	Do not literally throw away the data that you choose to exclude.
keep in mind that	remember this important point	Keep in mind that outliers do not necessarily represent an error.

Rules and explanations

The ideas

Figures add information rather than repeat it, are as simple as possible (colour only if necessary) and can be understood without the text.

- Graphs are slightly wider than tall; independent variable on the *x*-axis, dependent variable on the *y*-axis; values increase left to right and bottom to top; axes begin at zero; concise axis labels with units; legends inside the figure; one simple font of 8 to 14 point.
- Caption: *Figure 1*. (italic, numbered in order) + a brief description ending in a period + what the reader needs (abbreviations, units, what the error bars represent).
- **Bar graphs** compare means (categorical *x* variable). **Error bars** usually show one **standard error**; a difference between means greater than two standard errors is generally statistically significant, so readers can "see" significance.
- **Line graphs** suit a quantitative *x* variable with a few levels, or time (Carlson and Conard's name quartiles). **Scatterplots** show individuals, with no connecting lines; a **regression line** may be added.

Tables suit many means and standard deviations (MacDonald and Martineau's mood and self-esteem study): horizontal lines at the top, the bottom and under the headings; every column headed; numbered from 1 with a brief, clear title. A **correlation matrix** (McCabe et al.) fills in only half the table, with dashes on the diagonal. Do not repeat table values in the text; point out the main trends.

Preparing data. Remove identifying information; store data and consent forms separately and securely; keep backups. Check the raw data for missing, illegible or impossible answers (a "12" on a 1-to-10 scale); if the main variables are missing, you may have to exclude the participant, but **set the data aside**, keep notes and report it. Enter each item of a multiple-item measure separately and let the software combine them. Then check reliability (Cronbach's α or a split-half correlation) and describe each variable, asking what the numbers *mean*.

Outliers

- Data-entry error: correct it.

- Error, misunderstanding or lack of effort (a 3-minute reaction time when most took seconds): exclusion can be justified, using the same criteria for everyone and reporting the number excluded and the criteria.
- Possibly real (60 or 70 sexual partners in Brown and Sinclair's sample): use the median, or analyse with and without the outliers; if the results differ, report both.

Planned and exploratory analyses. A **planned analysis** tests your hypothesis; an **exploratory analysis** looks for unexpected patterns ("Go on a fishing expedition", Bem). Every extra unplanned analysis increases the chance that a chance pattern looks real (a Type I error), so keep the two apart in your report and treat exploratory results as ideas to replicate. Above all, understand the descriptive statistics first: M = 34.32 versus 21.45 with d = 1.31 already shows that a treatment worked.

Language focus

Reporting statistics in words and numbers (APA style)

Rule	Example from the book
Words for numbers below 10 that are not precise statistics; numerals for 10 and above	three conditions; 25 research methods students
Statistics in numerals, usually rounded to two decimal places	"2.00" rather than "two" or "2"
Narrative: the words *mean*, *standard deviation*	The mean age of the participants was 22.43 years with a standard deviation of 2.34.
Parentheses: the italic symbols *M*, *SD*, *r*	...stronger intentions to have unprotected sex (*M* = 4.05, *SD* = 2.32) than those in a positive mood (*M* = 2.15, *SD* = 2.27).
Parallel construction	The treatment group had a mean of 23.40 (*SD* = 9.33), while the control group had a mean of 20.87 (*SD* = 8.45).

Avoid: "...while 20.87 was the mean of the control group, which had a standard deviation of 8.45."
Exclusions are reported in the passive and past tense with the number and the criterion: "Data from two participants **were excluded** because..."

> **Note for Arabic speakers** - In formal writing *data* is plural: "the data **are** complete", "**these** data", "set **them** aside", as the book writes.

Practice

Part 1: Reading questions

1. When does APA style use words for numbers, and when numerals?
2. Why is the book's parallel example better than the nonparallel alternative?
3. What do error bars usually represent, and what can a reader conclude from them?
4. Why is only half of a correlation matrix filled in?
5. What should you do with data you decide to exclude, and why?

6. Why should results from exploratory analyses be viewed skeptically?

Part 2: Exercises from the book (section 12.5; Exercises 1 to 5 are in Lessons 27 and 28)

1. Practice: In a classic study, men and women rated the importance of physical attractiveness in both a short-term mate and a long-term mate (Buss & Schmitt, 1993). The means and standard deviations are as follows. Men / Short Term: $M = 5.67$, $SD = 2.34$; Men / Long Term: $M = 4.43$, $SD = 2.11$; Women / Short Term: $M = 5.67$, $SD = 2.48$; Women / Long Term: $M = 4.22$, $SD = 1.98$. Present these results (a) in writing, (b) in a figure, (c) in a table. (Sketch the figure and table on paper.)
2. Discussion: What are at least two reasonable ways to deal with each of the following outliers based on the discussion in this chapter? (a) A participant estimating ordinary people's heights estimates one woman's height to be "84 inches" tall. (b) In a study of memory for ordinary objects, one participant scores 0 out of 15. (c) In response to a question about how many "close friends" she has, one participant writes "32."

Part 3: Language exercise

1. Rewrite in parenthetical form: "The mean age of the participants was 22.43 years with a standard deviation of 2.34."
2. Rewrite in the narrative, without symbols: "...those in a negative mood ($M = 4.05$, $SD = 2.32$) than those in a positive mood ($M = 2.15$, $SD = 2.27$)."
3. Correct the APA errors: "The test-retest correlation was 0.958 for all forty participants in 3 groups."
4. Rewrite in the passive: "We excluded one participant's reaction times because she took 3 minutes to respond."

Speak with your tutor

- "Read me five statistics. I will write each in APA style, in the narrative and in parentheses; correct my form and parallel construction."
- "Describe a figure to me (axes, bars, caption); I will say which APA guidelines it breaks."
- "In my study, the wristband shows that one student in the nap condition did not sleep at all. Let me explain what I would do with these data; question my decision as a reviewer would."

Writing task

Your research project: results, part 3 (descriptive results in APA style). Use the dataset from Lessons 27 and 28 (**invented data for practice**). Nap condition, recall: 24, 19, 22, 26, 21, 23, 18, 21, 25, 21; minutes asleep: 70, 60, 80, 85, 55, 70, 75, 65, 70, 70. Wake condition, recall: 18, 15, 21, 16, 12, 19, 16, 22, 14, 17. Imagine that a 21st participant (wake condition) was excluded because she fell asleep during the rest period, a criterion set before data collection.

Write the descriptive part of your results section (120 to 180 words): the exclusion and the final sample, the descriptive statistics for both conditions, d and r from Lesson 28, and a reference to one

figure; then write the caption for a bar graph with standard-error bars. If you work on your own topic, use your own invented dataset with one invented exclusion.

Checklist:

- Did you give the number excluded, the criterion and the fact that it was set in advance?
- Are numbers below 10 that are not statistics written as words, and statistics as numerals with two decimal places?
- Did you use words in the narrative and italic symbols in parentheses?
- Are the two conditions described with parallel construction?
- Does the caption start with "*Figure 1*." and say what the error bars represent?
- Did you use the past tense and, for the exclusion, the passive?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "Convincing", the xkcd cartoon in section 12.3, on graphs that persuade: http://imgs.xkcd.com/comics/convincing.png

Answer key

Part 1

1. Words for numbers below 10 that are not precise statistical results; numerals for 10 and above; statistical results always in numerals.
2. It expresses comparable results in the same structure (group, mean, *SD* in parentheses), so readers can compare them easily.
3. Usually one standard error in each direction; a difference between means greater than two standard errors is generally statistically significant.
4. The other half would repeat the same values, and each variable's correlation with itself (1.00) is replaced by a dash.
5. Set them aside (never delete them) and keep notes on why, because you must report the exclusions and you or another researcher may want to see the data later.
6. Complex data contain chance patterns, and every unplanned analysis increases the risk of treating one as real (a Type I error); such results should be replicated before being presented.

Part 2 (example answers)

1. (a) "Men rated physical attractiveness as more important in a short-term mate (*M* = 5.67, *SD* = 2.34) than in a long-term mate (*M* = 4.43, *SD* = 2.11), and so did women (short term: *M* = 5.67, *SD* = 2.48; long term: *M* = 4.22, *SD* = 1.98)." (b) A bar graph: *x*-axis "Type of mate" (short term, long term) with a bar for men and a bar for women at each level and the legend inside; *y*-axis "Mean importance of physical attractiveness", starting at 0. Caption: "*Figure 1*. Mean importance of physical attractiveness in a short-term and a long-term mate, by sex. Error bars represent standard deviations." (Standard errors would need the sample sizes, which are not given.) (c) **Table 1**, title *Means and Standard Deviations of the Importance of Physical Attractiveness by Sex and Type of Mate*; columns Sex, Short-term *M* and *SD*, Long-term *M* and *SD*; rows Men 5.67, 2.34, 4.43, 2.11 and Women 5.67, 2.48, 4.22, 1.98.
2. Other reasonable ways exist. (a) Check the data file for an entry error (for example, 64 typed as 84) and correct it; if the participant really wrote 84, exclude it as a probable misunderstanding or lack of effort, by a criterion applied to everyone, and report it. (b) Check for a recording problem; exclude the participant if they probably misunderstood the task or did not try, and report the criterion; or keep the score and analyse with and without it. (c) It may be honest (people define "close friends" differently): keep it and use the median, or analyse with and without it.

Part 3

1. Participants had a mean age of 22.43 years (*SD* = 2.34).
2. ...those in a negative mood had a mean of 4.05 with a standard deviation of 2.32, while those in a positive mood had a mean of 2.15 with a standard deviation of 2.27.
3. The test-retest correlation was .96 for all 40 participants in three groups.
4. One participant's reaction times were excluded because she took 3 minutes to respond.

Writing task. Standard errors: nap 2.54 / √10 = 0.80; wake 3.09 / √10 = 0.98; the difference (5.00) is more than two standard errors. Example: "Data from one participant in the wake condition were excluded because she fell asleep during the rest period, a criterion set before data collection. The final sample consisted of 20 students, 10 per condition. Participants in the nap condition recalled more word pairs (M = 22.00, SD = 2.54) than participants in the wake condition (M = 17.00, SD = 3.09), d = 1.86 (see Figure 1). In the nap condition, minutes asleep (M = 70.00, SD = 8.82) correlated positively with recall, r = .45." Caption: "*Figure 1.* Mean number of word pairs recalled (out of 40) in the nap and wake conditions. Error bars represent standard errors."

My notes